Graphene through Chemical Route Technology

The Centre for Materials for Electronics Technology is a premier research centre in Electronic materials under Ministry of Electronics and Information Technology, Government of India. C-MET is engaged in the development of process/product/technology and transfer of the technology in the area of Electronic materials, components and devices to cater to India's strategic and industrial applications, exploiting indigenous resources of raw materials.

The C-MET family comprises three Centres, located at Pune, Hyderabad and Thrissur with specialized research mandate at each place. C-MET, Thrissur is concentrating on the development of electronic ceramic materials and devices. Its main areas of focus include nano-materials, nano-porous materials, sensors and actuators, microwave materials & nano-structured electronics.

Graphene based material is upcoming fields in electronics because of its prominent intrinsic properties, which may rival or even surpass both single and multiwalled carbon nanotubes. Graphene has good chemical stability, high electrical conductivity and large surface area and can be prepared from natural graphite flakes, which are available in plenty in nature which will make it a cost effective material if produced in bulk. Graphene is a promising electrode material in many applications which include battery, supercapacitors etc. Currently, more practical solution-chemistry based approaches involving the initial oxidation of graphite to graphite oxide, followed by exfoliation of graphite oxide to graphene oxide sheets, and then reduction to graphene have attracted much attention because of the facile scalability and high yields.

ABOUT THE PRODUCTS:

C-MET has developed a technology for the synthesis of graphene through chemical route from natural graphite flakes. India is having second largest deposition of natural graphite flakes and hence graphene can be made in a cost effective way. Using this graphene C-MET has developed supercapacitors, actuators and transparent electrodes, etc. We have made the plant design

including materials for construction and safety features for a plant of 100g capacity. C-MET can demonstrate 30g batch size at lab level. The major specification is

Type: Chemically reduced
Colour: Black
Number of layer: Few layer
Surface Area: BET Surface area 400 – 600 m^2/g (Varies with input natural graphite flakes)



Graphene developed through chemical route at C-MET

Technology Transfer

C-MET, Thrissur would like to invite competent industrial partners for commercialization of the above listed product through transfer of technology. The technology will be transferred on non-exclusive basis. The ToT package contains the following

1. Document for preparation process
2. Processing details
3. Product specification
4. Test plan and procedure
5. Training

The level of technology readiness at C-MET

The technology readiness level is 4

Period of training under the agreement

C-MET, Thrissur will give 2 months training to the engineers of the ToT partner at the premises of C-MET. 30g level batch will be demonstrated. 4 batches will be demonstrated.

Who can apply?

Industries/ start ups can approach C-MET, Thrissur for the technology. Professionally managed Companies and Corporates are also welcome to apply for the technology. A Committee of experts constituted by the C-MET will assess capabilities and strengths of the industry before finalizing the technology partners.

How to apply?

Interested companies may send expression of interest with their details by filling the following questionnaire along with supporting documents to:

Programme Co-ordinator
Centre for Materials for Electronics Technology (C-MET)
Panchawati, Off Pashan Road, Pune:411008
Email: prog.coord@cmet.gov.in

General:

The industry willing to take technology for commercial exploitation will be required to enter into a ToT agreement with C-MET, Thrissur as per the terms and conditions approved by the competent authority in the MeitY prescribed format.

ToT Cost:

The ToT cost for Graphene through Chemical Route Technology is Rs. 9.9 lakhs plus applicable taxes.